Total No. of Questions : 7] [Total No. of Printed Pages : 3

Roll No. ....

REF NO. Conf/Even/N/2025–26/20

B. Sc. (Hons) Semester IV Examination 2 0 2 5 – 2 6 (NEP)

BOTANY

Paper No. BOTMJ43 : Gymnosperms and Paleobotany

Time : 2½ hours *Full Marks* : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks

Answer **five** questions including Question **1** which is compulsory

1. Answer the following questions in not more than *100* words each : 7×2

   (a) Why are the leaves of *Cycas* described as pinnately compound but fern-like in appearance?

   (b) Describe the salient features of Progymnosperm.

(c) Why are the leaves of *Ephedra* reduced and what is the role of its green, jointed stems?

(d) Give an account of the affinities of *Pteridospermales* with *Cycadophyta* and ferns.

(e) Why are the seeds of *Pinus* winged and how are these wings derived?

(f) Describe the anatomical features of Lyginopteris.

(g) What is the significance of the presence of resin canals for survival in *Pinus* ?

2. Describe the geological time scale, outlining its major divisions, key events in each era and its significance in understanding the evolution of plant life. 14

3. Write short notes on *any two* of the following : 2×7

(a) Reproductive parts of Lyginopteris

(b) *Pteriodospermales*

(c) Classification of gymnosperms.

4. Give a comprehensive account of the types of fossils and their importance, with suitable diagrams. 14

5. Write short notes on *any two* of the following : 2×7

(a) Reproduction in *Pinus*

(b) Difference between young and old stems of *Ephedra*

(c) Structure and adaptations of *Pinus* leaf.

6. Describe the anatomy and reproduction of *Cycas*, highlighting its distinctive features and evolutionary significance. 14

7. Write short notes on *any two* of the following : 2×7

(a) Major geological events in Earth's history

(b) Glosspteris leaf

(c) Poikilohydry and homoiohydry.

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